

EEG-to-fMRI Brain Activation Prediction

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NE 422 – Final Report

Abstract

This project explores the feasibility of predicting functional magnetic resonance imaging (fMRI) brain activation maps from electroencephalography (EEG) recordings alone. Three deep learning architectures were implemented and evaluated: a compact convolutional neural network (EEGNet) paired with a multilayer perceptron (MLP) decoder, EEGNet paired with a U-Net decoder, and a cross-modal Transformer. All models were trained on simultaneous EEG-fMRI data recorded during motor execution and imagery tasks. While no model achieved a strongly positive correlation with ground-truth fMRI activation, the Transformer architecture yielded the most promising results (Pearson $r = 0.291$, SSIM = 0.443), demonstrating that EEG-derived signals carry some spatial information recoverable via learned embeddings.

1. Introduction

Electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) both serve as distinct windows into neural activity. EEG is inexpensive, portable, and boasts millisecond temporal resolution, but provides poor spatial localization. In contrast, fMRI measures blood-oxygen-level-dependent (BOLD) contrast with millimeter spatial precision but is expensive, requires the subject to remain stationary inside a large magnet, and is constrained by the hemodynamic response function to a temporal resolution on the order of seconds. A technique capable of inferring fMRI-quality spatial activation maps from EEG signals would have transformative implications for brain-computer interfaces (BCIs), clinical neuroimaging, and real-time brain state decoding.

The theoretical basis for such an approach rests on neurovascular coupling: the well-established relationship between neuronal electrical activity, as captured by EEG, and the local increase in cerebral blood flow that generates the BOLD signal. Prior work has begun to computationally explore this relationship. E2fNet (Roos, Fukuda, & Cap, 2025) demonstrated that a neural network trained on resting-state simultaneous EEG-fMRI data could synthesize plausible fMRI volumes from EEG, albeit with training datasets substantially larger than those available in many research contexts. The foundational Transformer architecture (Vaswani et al., 2017) introduced cross-attention mechanisms that have since enabled multimodal learning across diverse sensory domains, offering a natural framework for mapping EEG feature tokens to spatially indexed fMRI voxels. EEGNet (Lawhern et al., 2018) provides a compact convolutional encoder specifically designed for EEG signals, making it well-suited as a feature extractor in a pipeline targeting fMRI synthesis.

The dataset used in this project was collected by Bondi et al. (2025) and consists of simultaneous EEG-fMRI recordings from 17 healthy participants performing six motor conditions: right- and left-hand execution, right- and left-hand imagery, and right-foot execution and imagery. The original study used this dataset to investigate neurovascular coupling across motor conditions. Each subject contributed three runs, yielding approximately 460 fMRI volumes per run at a 2.0-second repetition time. EEG was recorded from 63 channels at 5000 Hz and preprocessed to 250 Hz with a 0.2–40 Hz bandpass filter.

The central hypothesis of this project is that EEG motor signals contain sufficient spatial information to approximate fMRI activation patterns for the same motor task. This hypothesis is tested by training models to regress directly from preprocessed EEG epochs to fMRI t-statistic maps derived from a general linear model (GLM) applied to the BOLD time series.

2. Methods

2.1 Preprocessing

EEG data were epoched around motor task onset events over a 0–3 second post-stimulus window using custom Python scripts built on MNE-Python and PyTorch dataloaders. Electrode Ch32, designated as an ECG artifact channel, was excluded, leaving 62 EEG channels. To capture temporal dynamics relevant to the hemodynamic response, a multi-lag representation was constructed using five lags spaced 2 seconds apart, yielding an input tensor of shape (62 channels $\times$ 5 lags $\times$ 751 timepoints). The lag spacing was chosen to approximately sample the rising phase and peak of the canonical hemodynamic response function, which peaks at approximately 5–6 seconds post-stimulus.

fMRI data were analyzed using a General Linear Model (GLM) implemented in Nilearn with a standard hemodynamic response function (SPM HRF). The GLM models the observed BOLD signal Y at each voxel as a linear combination of task regressors convolved with the HRF:

$$Y = X\beta + \varepsilon, \varepsilon \sim N(0, \sigma^2 I)$$

where $X \in \mathbb{R}^{(T \times K)}$ is the design matrix with T scans and K regressors, $\beta \in \mathbb{R}^K$ is the vector of estimated regression coefficients, and ε is the residual noise term. A high-pass filter cutoff of 1/128 Hz was applied to the BOLD time series to remove low-frequency drift prior to GLM fitting. Six condition regressors were included, corresponding to the motor conditions in Bondi et al. (2025): right- and left-hand execution, right- and left-hand imagery, and right- and left-foot execution and imagery.

To isolate lateralized motor cortex activation, differential contrasts (right-hand minus left-hand) were computed using contrast vector c , producing voxel-wise t-statistic maps:

$$t = c^T \hat{\beta} / \sqrt{(\sigma^2 c^T (X^T X)^{-1} c)}$$

These t-statistic maps served as the regression targets for all models. Subjects 1–13 were used for training, subjects 14–15 for validation, and subjects 16–17 for the held-out test set. Rest epochs were assigned zero activation maps, treating baseline periods as neurally silent, a simplification acknowledged in Section 4.

2.2 EEGNet + MLP

The first architecture combined the EEGNet convolutional encoder (Lawhern et al., 2018) with a fully connected multilayer perceptron (MLP) decoder, implemented in PyTorch. EEGNet applies a temporal convolution to learn frequency-band filters, followed by a depthwise spatial convolution analogous to independent component analysis for electrode weighting, and a separable convolution with average pooling to compress and denoise the representation. The

input tensor had dimensions $62 \text{ channels} \times 751 \text{ timepoints}$ per epoch. The resulting feature vector was passed to an MLP that independently predicted fMRI t-statistic values for all voxels in the target volume ($79 \times 95 \times 79$ voxels). Because each voxel is treated as a separate regression target with no explicit spatial structure, the MLP has no mechanism for enforcing anatomical coherence in predicted activation maps.

2.3 EEGNet + U-Net

The second architecture replaced the MLP decoder with a U-Net, also implemented in PyTorch. U-Net progressively upsamples the encoded feature representation back to the full fMRI volume resolution of $79 \times 95 \times 79$ voxels. Originally developed for biomedical image segmentation, U-Net is well-suited for volumetric reconstruction because skip connections between encoder and decoder pathways preserve spatial context, allowing neighboring voxels to be predicted jointly and promoting anatomical coherence. This model was expected to produce more spatially realistic activation maps than the MLP at the cost of higher model complexity and greater demand for training data.

2.4 Cross-Modal Transformer

The third architecture was a cross-modal Transformer with 6.4 million parameters, implemented in PyTorch following the framework of Vaswani et al. (2017). EEG data were decomposed into frequency bands and embedded into 256-dimensional tokens augmented with positional embeddings encoding temporal information and channel embeddings encoding electrode identity. Four Transformer encoder layers applied multi-head self-attention across EEG tokens to produce a sequence of EEG feature representations $H \in \mathbb{R}^{(N \times d)}$, where N is the number of tokens and $d = 256$ is the embedding dimension.

On the decoder side, a VoxelPositionalEmbedding module encoded the three-dimensional MNI coordinates of each fMRI voxel, normalizing coordinates and projecting them through an MLP to produce spatially-aware query vectors $Q \in \mathbb{R}^{(V \times d)}$, where V is the number of voxels. Four cross-attention decoder layers allowed each voxel query to attend to the full sequence of EEG feature tokens via scaled dot-product cross-attention:

$$Attention(Q, K, V) = softmax(QK^T / \sqrt{d_k}) V \quad (3)$$

where $Q = W_Q \cdot q_{\text{voxel}}$ are the voxel-derived queries, $K = W_K \cdot H$ and $V = W_V \cdot H$ are the EEG-derived keys and values, d_k is the key dimension, and W_Q, W_K, W_V are learned projection matrices. This mechanism allows each spatial location to selectively attend to the EEG features most predictive of its activation, providing an explicit inductive bias linking EEG spectral content to spatial fMRI patterns.

The model was trained using a composite loss function combining weighted mean squared error, Pearson correlation, and a variance-matching penalty:

$$L = \alpha \cdot MSE(\hat{y}, y) - \beta \cdot r(\hat{y}, y) + \gamma \cdot |Var(\hat{y}) - Var(y)| \quad (4)$$

where $\hat{y}$ and y are predicted, and ground-truth t-statistic maps, $r(\cdot, \cdot)$ denotes Pearson correlation, and α, β, γ are scalar weights tuned during validation. The variance-matching term was included

to discourage mean-collapse behavior. Training used the Adam optimizer with a batch size of 4 over 6 epochs, limited by available compute resources (laptop GPUs).

2.5 Alternative Approaches and Challenges

Several implementation challenges were encountered during development. The MLP decoder exhibited mean collapse, converging to predict a near-constant average activation volume regardless of EEG input. This reflects insufficient model capacity or training signal to capture subject- and trial-level variability in the low-data regime. The composite loss in Equation (4) was introduced specifically to counteract this tendency in the Transformer, with the correlation and variance terms penalizing flat predictions.

U-Net training produced more spatially diverse predictions than the MLP, but was hampered by dataset size. The spatial upsampling pathway requires substantially more paired examples to learn anatomically coherent mappings, and with only 13 training subjects, the U-Net overfitted to subject-specific noise rather than generalizing to motor cortex structure.

A two-stage approach was considered in which EEG first predicts the motor condition label via a classifier, which then indexes a condition-specific fMRI activation template. This would reduce the problem from direct voxel regression to template retrieval — a substantially simpler learning objective — but was not implemented due to time constraints. Future work should evaluate this pipeline as an intermediate step toward full voxel-level synthesis.

3. Results

Performance metrics for all three models on the test set (subjects 16–17) are summarized in Table 1. The EEGNet + MLP model achieved a mean squared error (MSE) of 0.740 and a mean absolute error (MAE) of 0.424, but a negative R^2 of -0.226 and a Pearson correlation of 0.0338 with ground-truth t-statistic maps, indicating that predictions were not meaningfully better than a constant predictor. The EEGNet + U-Net model performed worse on MSE (0.960) and correlation (0.00256), with $R^2 = -0.589$, suggesting that the added spatial complexity of the U-Net decoder was detrimental without the necessary amount of training data to properly train it. The Transformer model recorded the highest MSE (1.04) and MAE (0.608), but achieved substantially higher Pearson correlation ($r = 0.291$) and a Structural Similarity Index (SSIM) of 0.443 (mean per-slice SSIM = 0.484), indicating that the Transformer predictions captured more of the spatial structure of real activation maps despite higher voxel-level error.

Table 1: Model Performance on Test Set (Subjects 16–17)

Model	MSE	MAE	R^2	Pearson r	SSIM
EEGNet + MLP	0.740	0.424	-0.226	0.034	—
EEGNET + U-Net	0.960	0.467	-0.589	0.003	—
Transformer	1.040	0.608	-0.406	0.291	0.443

Table 1. Test-set performance metrics for all three architectures.

Qualitative inspection of predicted activation maps (Figures 1–3) revealed that the MLP decoder produced spatially homogeneous and dampened predictions with little resemblance to ground-truth lateralized motor cortex activation, characteristic of mean-collapse failure. U-Net predictions showed greater spatial variation but only little correspondence to true activation regions. The Transformer produced maps with broader regional activation patterns that partially overlapped with the expected motor cortex and supplementary motor area activations, consistent with its improved correlation metric.

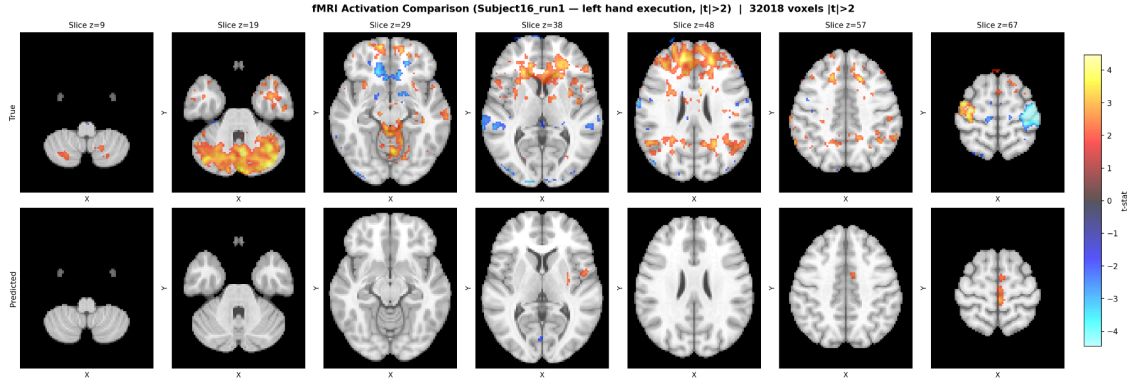


Figure 1. Predicted versus ground-truth fMRI t -statistic maps for a representative test subject using the EEGNet + MLP architecture. The MLP decoder produces spatially homogeneous predictions with minimal correspondence to the lateralized motor cortex activation observed in the ground-truth maps.

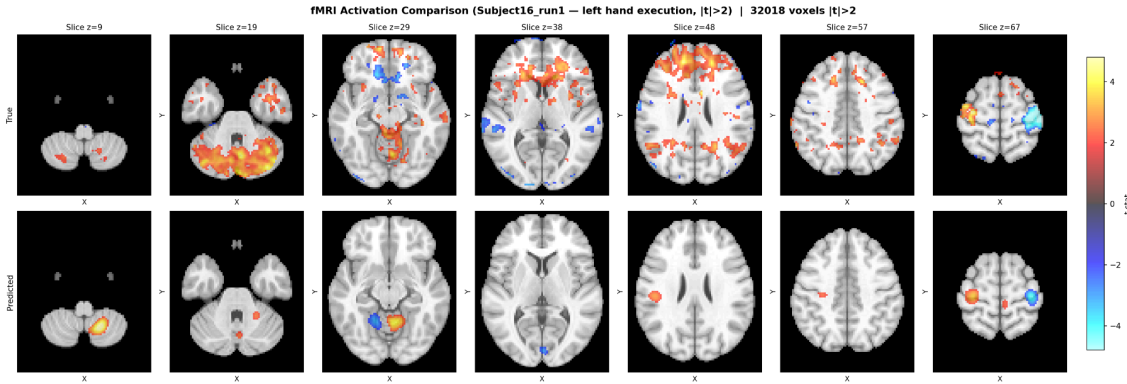


Figure 2. Predicted versus ground-truth fMRI t -statistic maps for a representative test subject using the EEGNet + U-Net architecture. While U-Net predictions exhibit greater spatial variation than the MLP, they do not reliably capture the spatial distribution of true motor activation.

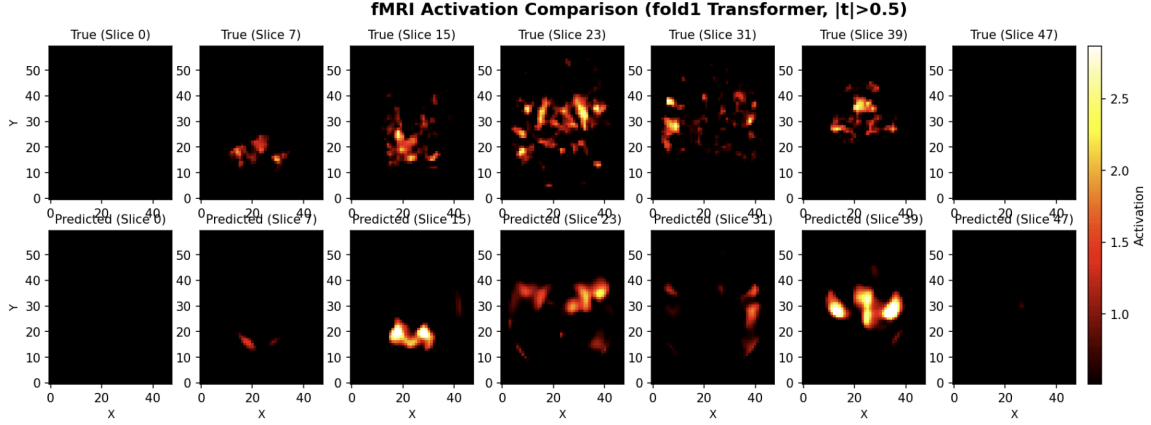


Figure 3. Predicted versus ground-truth fMRI t -statistic maps for a representative test subject using the cross-modal Transformer. Regional activation patterns show partial overlap with expected motor cortex and supplementary motor area activations, consistent with the higher Pearson correlation ($r = 0.291$) and SSIM (0.443) achieved by this model.

4. Discussion

The hypothesis that EEG motor signals contain sufficient spatial information to approximate fMRI activation maps was only weakly supported by the current results, given the current dataset size. The transformer architecture provided the most encouraging evidence, with a Pearson correlation of 0.291 suggesting that cross-modal attention can begin to extract spatially relevant features from EEG. No model achieved a positive R^2 , meaning no model consistently outperformed a simple mean-map baseline. Several factors likely contributed to this outcome.

Comparison with E2fNet (Roos et al., 2025) highlights the role of dataset size. E2fNet was trained on 14,688 paired samples from resting-state paradigms using the raw BOLD signal as the prediction target, preserving one sample per TR. In contrast, our approach computed a GLM across multiple trials per condition to produce t -statistic contrast maps as targets, which substantially reduced the number of usable paired samples; 13 subjects across 3 runs and 6 conditions give far fewer observations than a TR-by-TR pairing scheme. Additionally, our training set performed motor tasks across 6 conditions, substantially increasing trial-to-trial variability. Motor imagery and execution also produce fundamentally less stereotyped brain responses than resting state, making the EEG-to-fMRI mapping more complex and the learned representations less transferable across subjects.

The MLP mean-collapse failure is consistent with findings in low-data multimodal regression settings, where the model minimizes expected loss by predicting the conditional mean of the output distribution. The U-Net underperformed the MLP despite its architectural advantages, likely because the U-Net's spatial upsampling requires considerably more paired examples to learn anatomically coherent mappings. The Transformer's cross-attention mechanism, which explicitly queries EEG feature tokens at each voxel location, appears better suited to small-data cross-modal problems because it imposes stronger inductive biases about the structure of the EEG-fMRI relationship.

These results have implications for the neuroimaging field. They suggest that with larger datasets and refined training frameworks, learned EEG-to-fMRI mappings may be practical. Applications

could include low-cost approximations of fMRI for BCI decoding of motor intent, screening tools for conditions identified by motor cortex hypo- or hyperactivation, and real-time neurofeedback systems that use EEG to infer spatial brain state.

Limitations include the small training set, the treatment of rest epochs as zero-activation maps (which likely introduced confounds), hardware constraints limiting training to 6 epochs for the Transformer, and restriction to motor task conditions. The cross-subject generalization test (training on subjects 1–13, testing on subjects 16–17) is a demanding evaluation, and within-subject or subject-adaptive models may perform substantially better.

Future work should explore leave-one-subject-out cross-validation to maximize training data utility, subject-adaptive fine-tuning of pretrained models, and incorporation of larger simultaneous EEG-fMRI datasets. A two-stage prediction pipeline, in which EEG first predicts the motor condition label and then indexes a learned fMRI template, may provide a more tractable intermediate objective. Replacing MSE with SSIM as the primary training loss may better align optimization with perceptually meaningful spatial similarity.

Author Contributions

Bernie Dohrn: Transformer model design and implementation. Jack Dohrn: Preprocessing pipeline, EEGNet + MLP and EEGNet + U-Net implementation. John Parquette: Presentation and Paper outlines and writing. All authors contributed to experimental design, interpretation of results, and revision of the final report.

Code Availability

All code for preprocessing, model implementation, and training is available at the project [\[GitHub repository\]](#)

References

- Bondi, E., Ding, Y., Zhang, Y., Maggioni, E., & He, B. (2025). Investigating the neurovascular coupling across multiple motor execution and imagery conditions: A whole-brain EEG-informed fMRI analysis. *NeuroImage*, 121311. <https://doi.org/10.1016/j.neuroimage.2025.121311>
- Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018). EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces. *Journal of Neural Engineering*, 15(5), 056013. <https://doi.org/10.1088/1741-2552/aace8c>
- Roos, K., Fukuda, A., & Cap, Q. H. (2025). From brainwaves to brain scans: A robust neural network for EEG-to-fMRI synthesis. *arXiv preprint arXiv:2502.08025*.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. <https://arxiv.org/pdf/1706.03762>